

# SUNIX SDC Expansion Board Linux Library

## Digital I/O

### Development Instruction

Driver Ver. 2.1.1.0

Doc Ver. 2.0

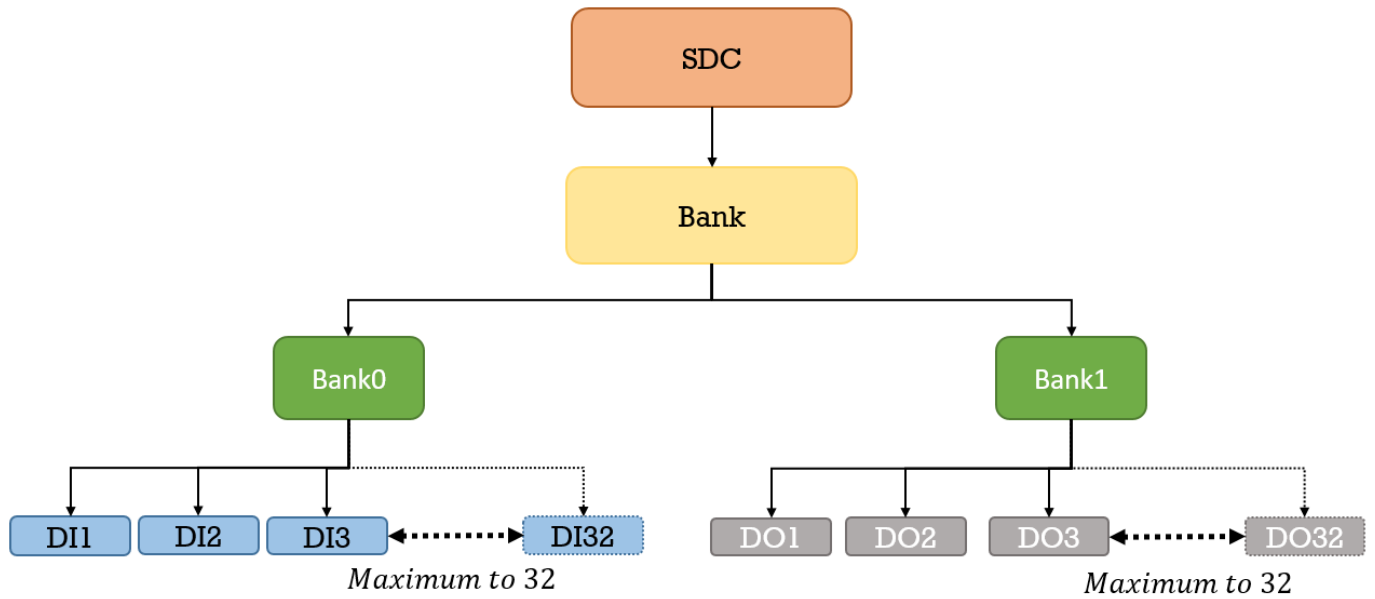
2024, August



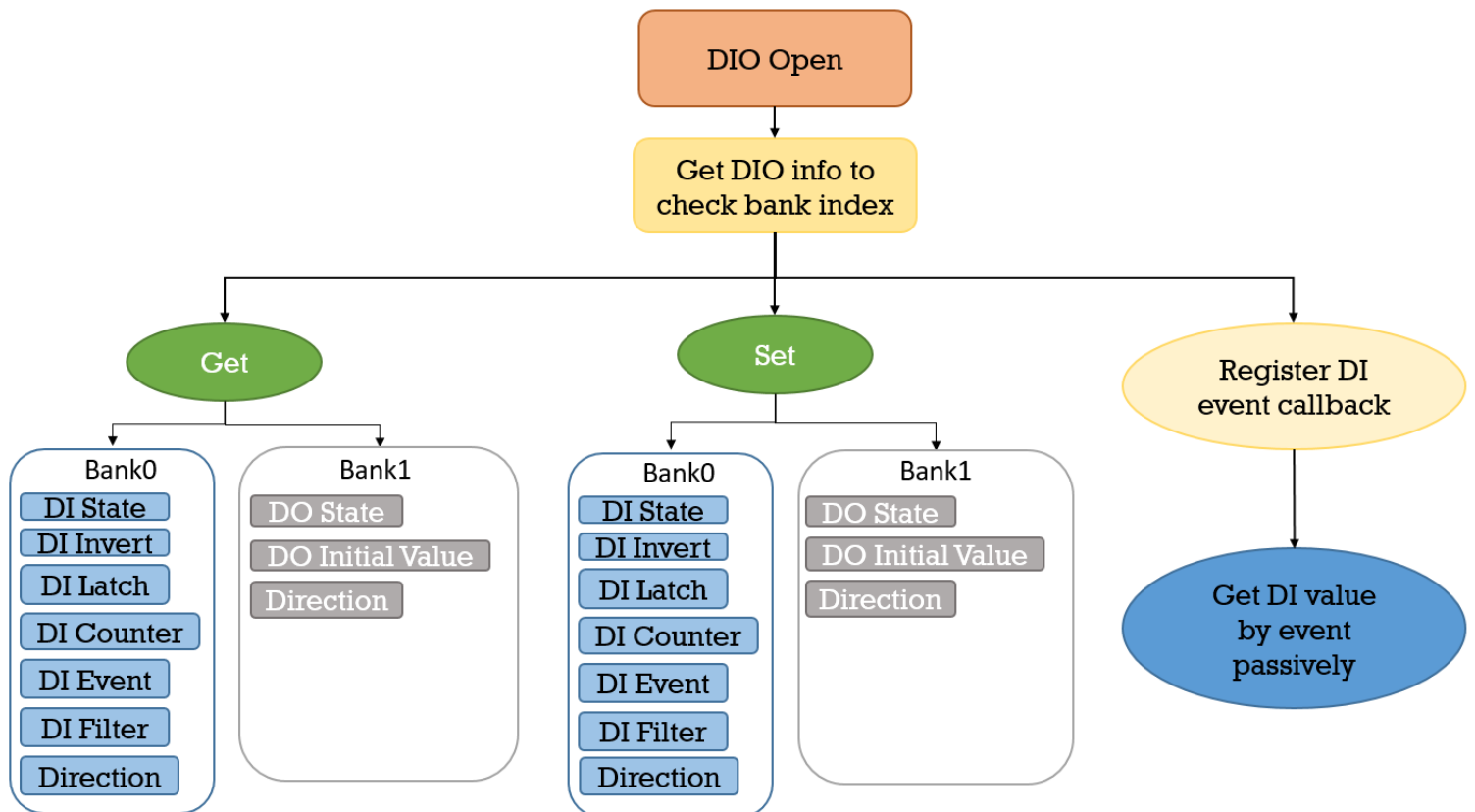
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## 1. SDC DIO layer



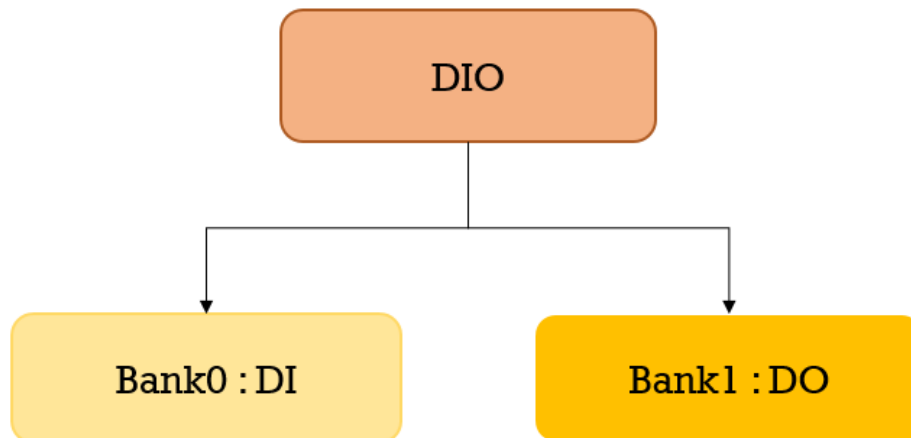
## 2. DIO program instruction



SDC library must to open the I/O channel first and then control the I/O.

There are two type of DIO Bank – DI and DO, if  $\text{cap\_input} = 1$ ; the bank is DI, if  $\text{cap\_output} = 1$ ; the bank is DO, about the IO amount of the bank, please check `NumberOfIO`

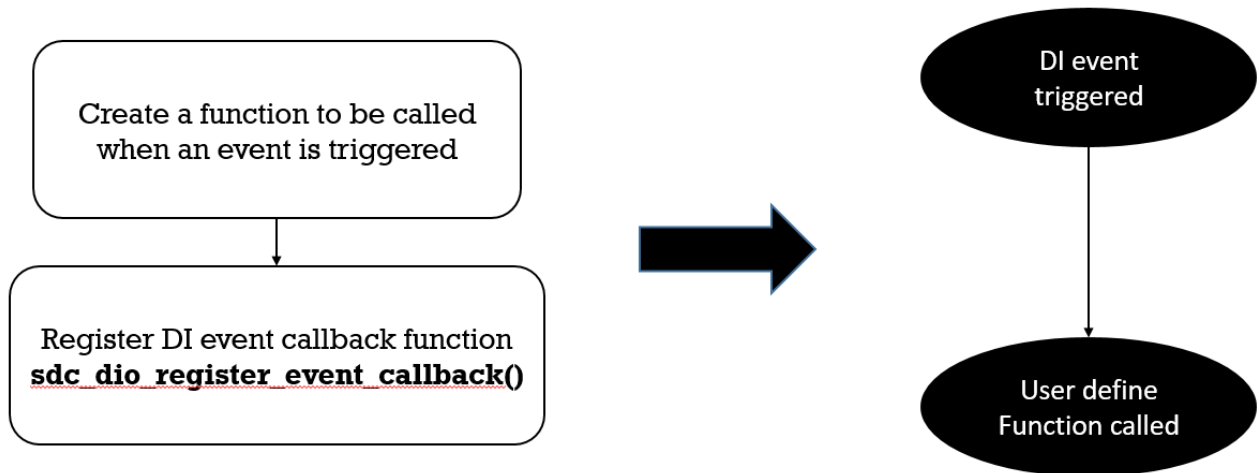
E.g. if  $\text{nr\_port} = 8, \text{cap\_input} = 1 \rightarrow$  the bank is 8DI group, Currently, All SDC DIO series product are **Bank0:DI**; **Bank1:DO**. These parameters defined in **`sdc_dio_info`** and **`sdc_dio_bank_info`** structures.



There are two mechanisms for DIO control – “get” and “set”, develop could get and set configuration and value by each relevant function. The following rules:

- A. Can only control all DI and DO pin (INT32 value), can’t control single pin.
- B. DI can only read and not write, if user write DI bank, would return fail.
- C. If user set DI configuration to DO bank, e.g. DI invert, would return fail.
- D. Current SDC series product doesn’t provide the function to setting bank direction.
- E.  $\text{DI Counter Value} = \text{Positive Counter} + \text{Negative Counter}$ .

Different from the way of actively requesting DI state, there is a special function about SDC API – DI event. Development could register and set a DI event callback function to get the DI state passively, before register and set DI event callback, developer must be set DI rising or falling event.



- About SDC API functions and structures, please refer to the document:  
“SUNIX IO Expansion Card Linux Library Development Manual.pdf”

### 3. Sample program installation and operation

#### 3.1. C/C++

- 1). First, Copy **libsdcio.so.x.x.x** to **/usr/lib**

The .so file position is:

**./SDCLinuxExpansionBoardDriverVX.X.X.X/lib/files/x86-64** or

**./SDCLinuxExpansionBoardDriverVX.X.X.X/lib/files/arm64**

```
$cp libsdc.so.x.x.x /usr/lib
```

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /lib/files/x86-64# ls
libsdc.so.2.0.0
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /lib/files/x86-64# cp libsdc.so.2.0.0 /usr/lib
```

- 2). Enter **ldconfig**.

```
$ldconfig
```

- 3). Enter C sample directory, compile it and execute it.

Directory: **./SDCLinuxExpansionBoardDriverVX.X.X.X/lib/sample\_code/sdc\_libtest**

\$make

\$/obj/sdclibtest

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_/lib/sample_code/c/sdc_libtest# ls
dio_callback.c dio_test.c function.h main.c Makefile obj precomp.h sdclib.h utils.c
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_/lib/sample_code/c/sdc_libtest# make
gcc -c -O2 -Wall dio_callback.c
gcc -c -O2 -Wall dio_test.c
gcc -c -O2 -Wall utils.c
gcc -c -O2 -Wall main.c
mv -f *.o obj
gcc -o sdclibtest \
obj/dio_callback.o \
obj/dio_test.o \
obj/utils.o \
obj/main.o \
-L obj -l:libsdc.so.2
rm -f obj/*.o
mv -f sdclibtest obj

root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_/lib/sample_code/c/sdc_libtest# ./obj/sdclibtest
```

- 4). Press 'E' key to enter DIO control list.

```
>-----<
E : DIO test
Q : quit
>-----<

> CMD ?
```

- 5). Press "B" key to open DIO channel.

```
>>-----<<
B : open
C : close
D : get and print info
F : get/set BANK direction
G : get/set BANK state
H : get/set BANK input invert
I : get/set BANK input latch positive edge
J : get/set BANK input latch negative edge
K : get/set BANK input counter increment positive edge
L : get/set BANK input counter increment negative edge
M : get/set BANK input event ctrl
N : get/set BANK output initial value
O : set BANK input counter reset
P : get BANK input counter value
R : get/set BANK input PORT filter value
8 : event callback register
9 : event callback unregister

? : Command list
Q : quit
>>-----<<

>> CMD ?

DIO line number ? 0
Open line(0), status:0
```

```
mem      nvme0n1p2  rtc
mqueue   nvram    rtc0
mtd0      port      sdc_config0
mtd0ro    ppp        sdc_dio0
mtd1      pps0     shm
mtd1ro    psaux    snapshot
net        ptmx     snd
ng0n1     ptp0     stderr
null       ptp1     stdin
nvme0      pts      stdout
nvme0n1    random  tpm0
nvme0n1p1  rfkill   tpmrm0
```

- 6). Press "D" key to get DIO Info, according SDC device node: /dev/config0, the line number is 0.  
Bank index 0 is input; Bank index 1 is output.

```
-----
pci_bus          : 4
irq              : 17
line             : 0
index            : 1
version          : 1
cap_samedirection : 1
cap_storeflash   : 1
nr_bank          : 2
di_sampling_freq : 500000
di_filter_lower_bound_ms : 1
di_filter_min_ms : 1
di_filter_max_ms : 10000
=====Bank0=====
nr_port          : 8
cap_input        : 1
cap_output       : 0
cap_rising_trigger : 1
cap_falling_trigger : 1
=====Bank1=====
nr_port          : 8
cap_input        : 0
cap_output       : 1
cap_rising_trigger : 0
cap_falling_trigger : 0
=====
-----
```

- 7). Press “G” key to get/set Bank state, get DI state, the bank index is 0x00, set DO state, the bank index is 0x01, There are 8DI/8DO in the model, that’s mean the DIO state range is 0x00000000 ~ 0x000000FF.

DIO Connection mode  
: Loopback

DO PWR ↔ DI COM

DO GND ↔ DI GND

DO1 ↔ DI1

DO2 ↔ DI2

DO3 ↔ DI3

DO4 ↔ DI4

DO5 ↔ DI5

DO6 ↔ DI6

DO7 ↔ DI7

DO8 ↔ DI8

```
Operation (0->get, 1->set) ? 0
Bank index                ? 0

-----
Get state ok
BANK                      : 0
Value                     : x00000000
-----
```

Get DI state = 0x00000000

```
Operation (0->get, 1->set) ? 1
Bank index                ? 1
Value (4 byte)            ? x000000ff
```

```
-----
Set state ok
BANK                      : 1
Value                     : x000000FF
-----
```

Set DO state = 0x000000FF

>> CMD ?

```
Operation (0->get, 1->set) ? 0
Bank index                ? 0
```

```
-----
Get state ok
BANK                      : 0
Value                     : x000000FF
-----
```

Get DI state = 0x000000FF

>> CMD ?

- 8). Press 'M' key to get/set DI rising/falling edge event. Rising and falling event enable.

```
Operation (0->get, 1->set)      ? 1
Bank index                     ? 0
Rising event ctrl (4 byte)     ? x000000FF
Falling event ctrl (4 byte)    ? x000000FF

-----
Set input event ctrl ok
RISING - Value                 : x000000FF
FALLING - Value                 : x000000FF
-----
```

Set all DI rising/falling edge event

- 9). Press "8" key to register DI event callback

```
>> CMD ?

-----
event callback register ok
-----
```

- 10). Control the DO to change DI state, received the event

```
>> CMD ?

Operation (0->get, 1->set)      ? 1
Bank index                     ? 1
Value (4 byte)                 ? x000000FF

-----
Set state ok
BANK                           : 1
Value                           : x000000FF
-----

>> CMD ?
#####
line                           : 0
BANK                           : 0
INPUT                          : x000000FF
INPUT_DELTA                     : x000000FF
#####
```

Set DO state = 0x000000FF

Passively receive DI event



## 3.2. Python

### 1). Install python3

```
$apt-get install python3
```

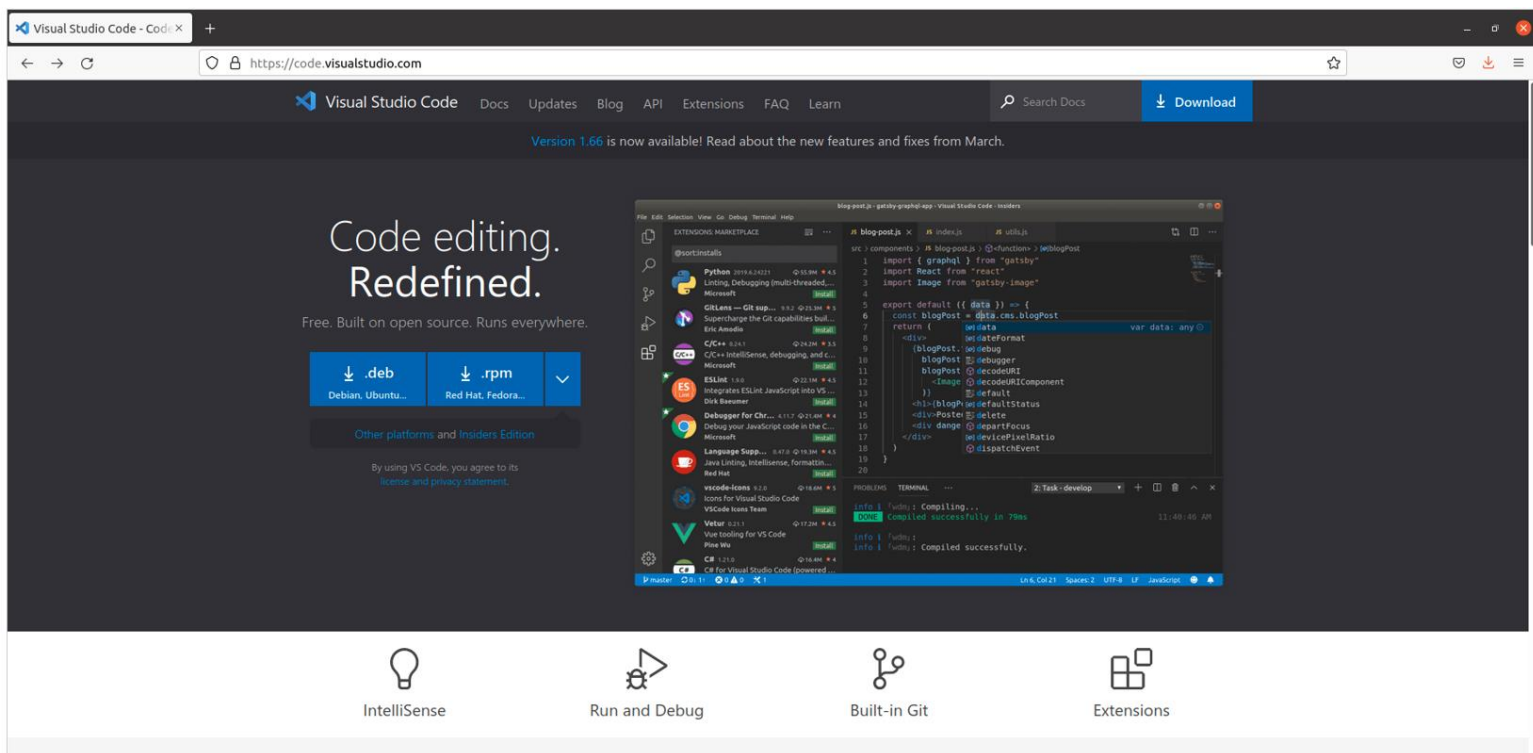
### 2). Install pip

```
$apt-get install python3-pip
```

### 3). Install click module by pip

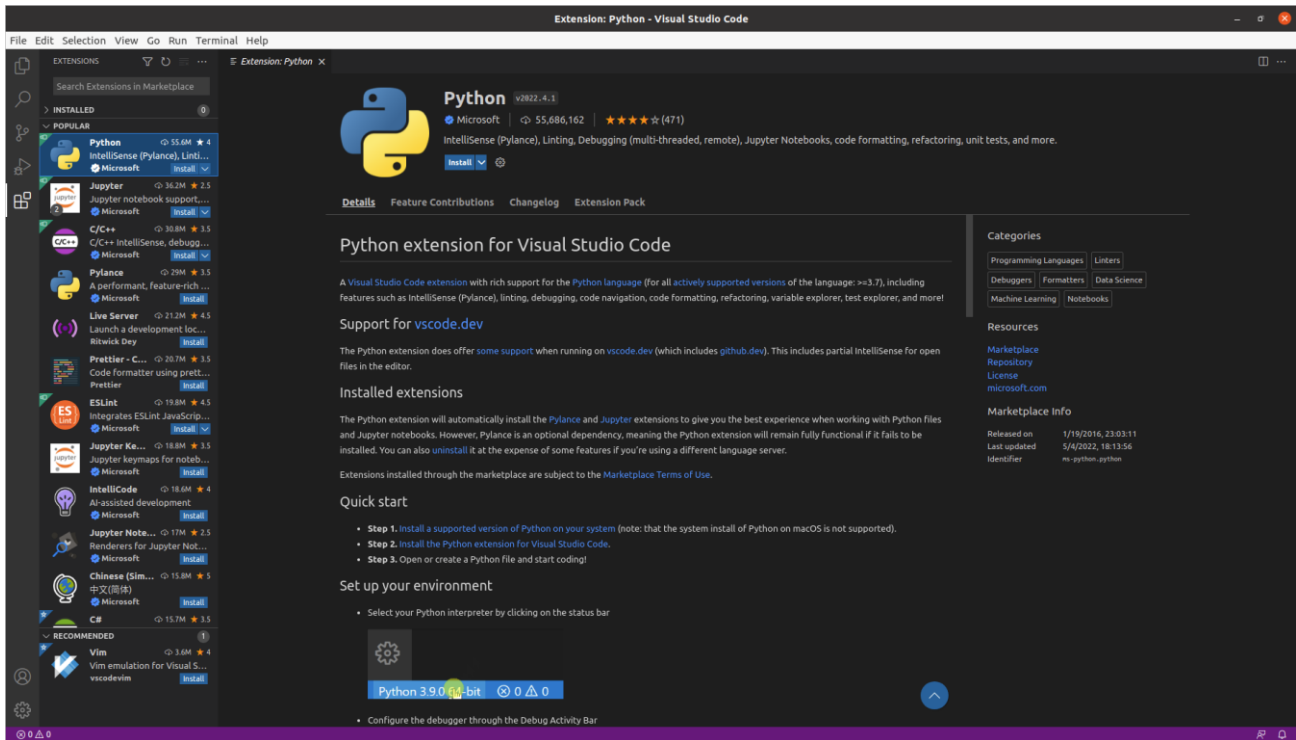
```
$pip3 install click
```

### 4). Install IDE, This case use Visual Studio Code as example, developer could download the deb file by official website and install it.



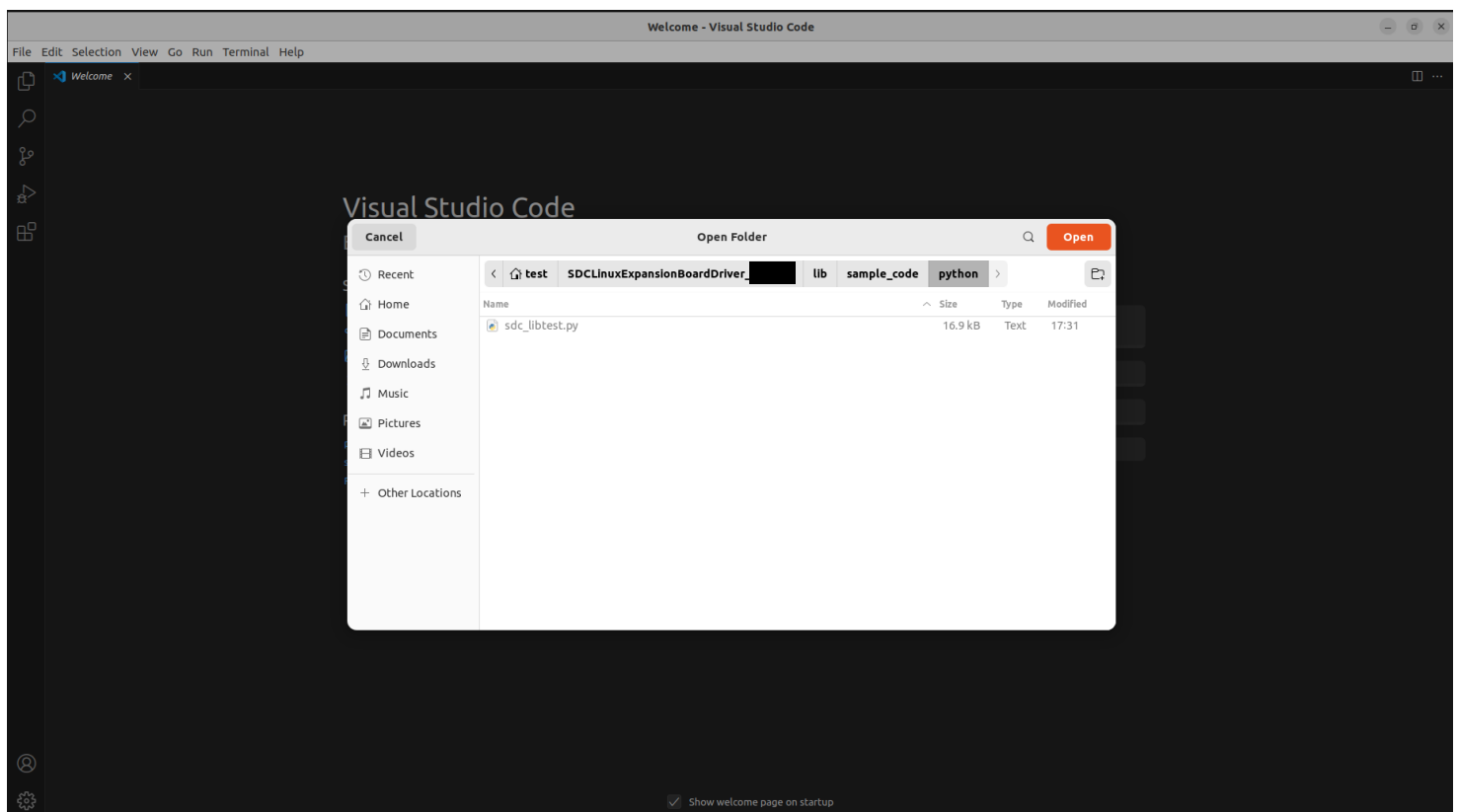
Anyway, developer can choose IDE according to your preferences, e.g. PyCharm, Spyder, Atom, Eclipse, or even vim.

### 5). Install the Python development kit for VScode

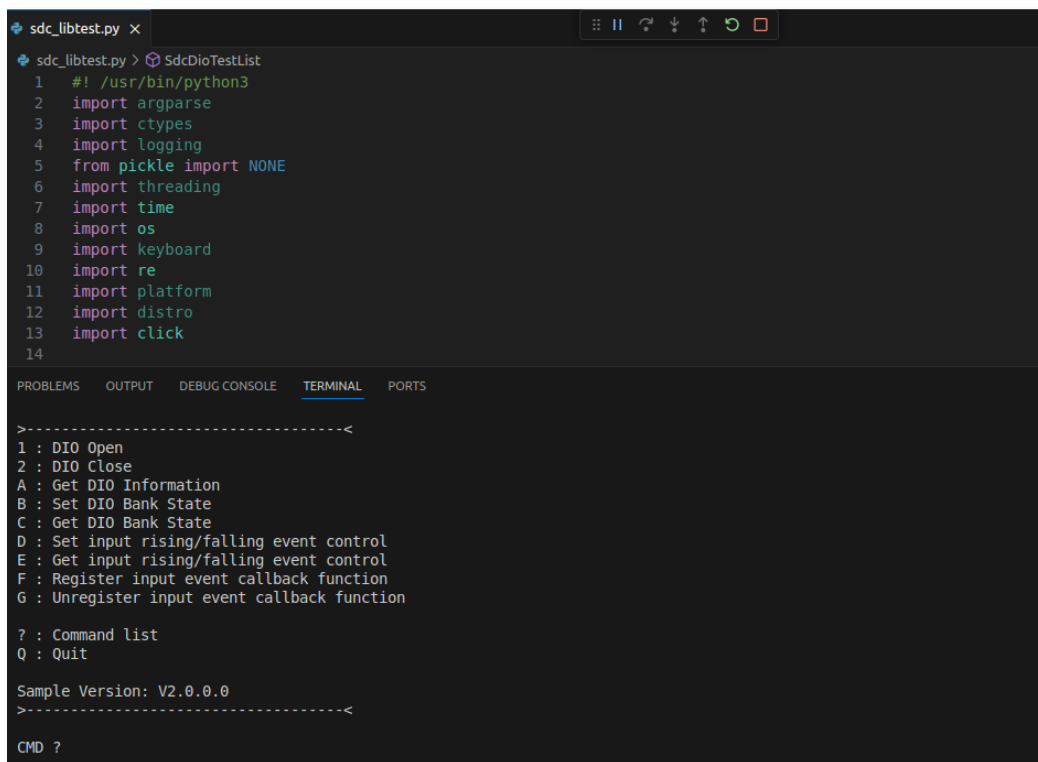
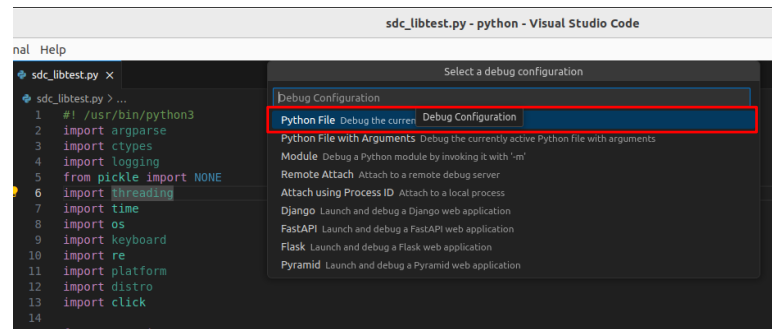
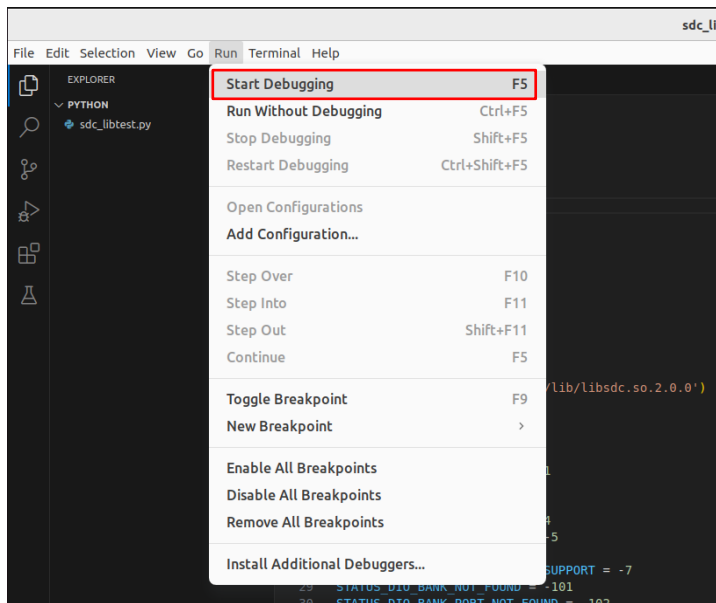


- 6). Open the folder which content the sdctest.py sample  
The directory:

`./SDCLinuxExpansionBoardDriverVX.X.X.X/lib/sample_code/python`



## 7). Start to debug sdctest.py



## 8). Stop VSCode debug, run sdctest.py on terminal

`./sdctest.py`

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /lib/sample_code/python# ls
sdctest.py
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /lib/sample_code/python# ./sdctest.py
```

```
>-----<
1 : DIO Open
2 : DIO Close
A : Get DIO Information
B : Set DIO Bank State
C : Get DIO Bank State
D : Set input rising/falling event control
E : Get input rising/falling event control
F : Register input event callback function
G : Unregister input event callback function

? : Command list
Q : Quit

Sample Version: V2.0.0.0
>-----<

CMD ?
_
```

9). Press “1” key to open DIO

```
CMD ?
line?(positive integer):
0
sdc_dio_open return code:0

#####
sdc_dio_open ok
#####

CMD ?
_
```

10). Press “A” key to Get DIO Info

```
CMD ?
line?(positive integer):
0
sdc_dio_get_info return code:0
#####
PCI Bus Number: 4
IRQ: 17
Line: 0
Index: 1
Version: 1
Same direction capability: 1
Store flash capability: 1
Number of bank: 2
DI sampling frequency(Hz): 500000
DI sampling lower bound value(ms): 1
DI sampling min value(ms): 1
DI sampling max value(ms): 10000
=====Bank:0=====
DIO Amount:8
DIO Type: Input
DI event rising trigger capability:1
DI event falling trigger capability:1
=====Bank:1=====
DIO Amount:8
DIO Type: Output
DI event rising trigger capability:0
DI event falling trigger capability:0
#####
```

- 11). Press “B” and “C” key to set DO state, get DI state, the bank index is 0x00, set DO state, the bank index is 0x01, There are 8DI/8DO in the model, that’s mean the DIO state range is 0x00000000 ~ 0x000000FF.

DIO Connection mode

: Loopback

DO PWR ↔ DI COM

DO GND ↔ DI GND

DO1 ↔ DI1

DO2 ↔ DI2

DO3 ↔ DI3

DO4 ↔ DI4

DO5 ↔ DI5

DO6 ↔ DI6

DO7 ↔ DI7

DO8 ↔ DI8

```
CMD ?
line?(positive integer):
0
sdc_dio_open return code:0
```

```
#####
sdc_dio_open ok
#####
```

```
CMD ?
line?(positive integer):
0
Bank Index?( 0:input / 1:output )
0
#Get Bank:0 State:0x00000000
```

DI state = 0x00000000

```
CMD ?
line?(positive integer):
0
Bank Index?( 0:input / 1:output )
1
State?(4 byte hex value, e.g. 000000FF):
000000ff
#Set Bank:1 State:0x000000ff
```

Set DO = 0x000000FF

```
CMD ?
line?(positive integer):
0
Bank Index?( 0:input / 1:output )
0
#Get Bank:0 State:0x000000ff
```

DI state = 0x000000FF

- 12). Press ‘D’ key to set DI rising/falling edge event.

```
CMD ?
line?(positive integer):
0
Bank Index?(0:input / 1:output):
0
Rising edge event?(4 byte hex value, e.g. 000000FF):
000000ff
Falling edge event?(4 byte hex value, e.g. 000000FF):
000000ff
#####
#Set Bank:0 Rising edge event:0x000000ff
#Set Bank:0 Falling edge event:0x000000ff
#####
```

Set all DI  
rising/falling edge  
event

- 13). Press “F” key to register DI event callback

```
CMD ?
line?(positive integer):
0
#####
sdc_dio_register_event_callback ok
#####
```

14). Control the DO to change DI state, received the event

```
CMD ?
line?(positive integer):
0
Bank Index?( 0:input / 1:output )
1
State?(4 byte hex value, e.g. 000000FF):
000000ff
#Set Bank:1 State:0x000000ff
***** Input Event Triggered *****
Line:0
Bank Index:0
Input Delta:0x000000ff
Input Value:0x000000ff
*****
```

DI state = 0x00000000

Passively receive DI event